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It was never the R score.

How a scholarship committee penalized remote regions twice, and a model that removes both biases

The gap the audit found

Montréal + Capitale-Nationale

48.4%

granted · average R score 28.0

Bas-Saint-Laurent, Côte-Nord, Gaspésie

27.3%

granted · average R score 27.3

A 0.7-point R-score difference cannot explain a 21-point gap.

Same R score, different outcome

R-score band	Centres	Remote	Gap
26 – 27	16%	4%	12 pts
27 – 28	32%	10%	22 pts
28 – 29	59%	26%	34 pts
29 – 30	85%	57%	28 pts
30 – 32	96%	83%	13 pts

At equal performance

34-point gap

for students right at the cutoff

We rebuilt the committee's rule

$1.39 \times R$

Merit

academic performance

$+0.20 \times \text{hours}$

Merit

working while studying

**$+1.78 \times \log$
income**

Bias: wealth bonus

doubling income $\approx +0.9$ R pts

$-2.15 \times \text{remote}$

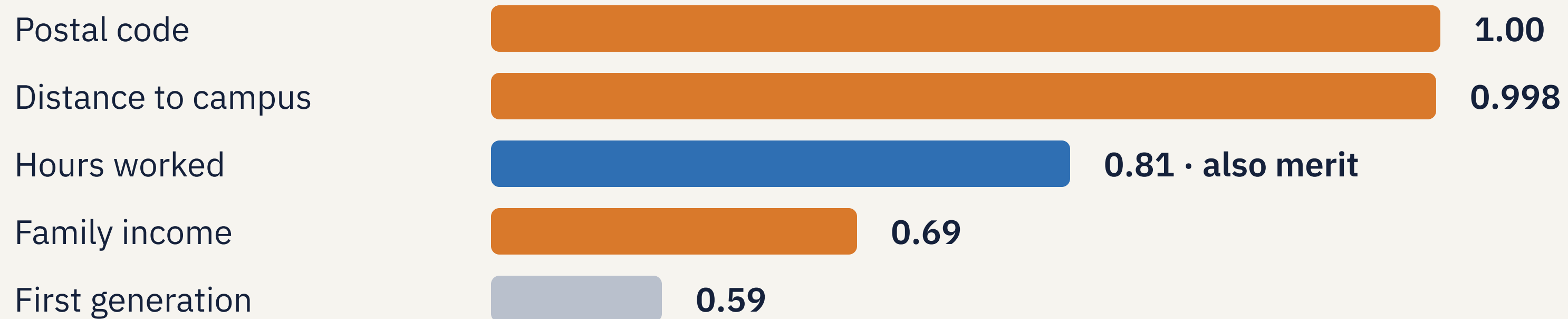
Bias: regional penalty

≈ -1.6 R pts

Remove both bias terms: remote grant rate 27% $\rightarrow$ 49.5%. Centres: 48.4%.

The two terms explain the whole gap. First generation, distance and program: no effect.

Deleting the region column doesn't work



The right question isn't "does it correlate with region?" but "does it measure merit?"

THE KEY FINDING

Once hours count, both groups are equally meritorious: 39.7% vs 39.8%.

Remote students' slightly lower R scores are offset by 4 more hours of work per week. **The entire 21-point gap is bias.**

Our metric: **equal opportunity, measured against merit**, not against the committee's labels. Production model gap: 0.29 (official figure: 0.270).

Our fix: show the bias to the model, then switch it off

1

Train

Logistic regression on the committee's decisions, with region, income, distance, first generation and program included as context, so they absorb the bias.

2

Neutralize

At prediction time, every candidate gets the same context. The model learns the hours weight itself: 0.146 R pts per hour.

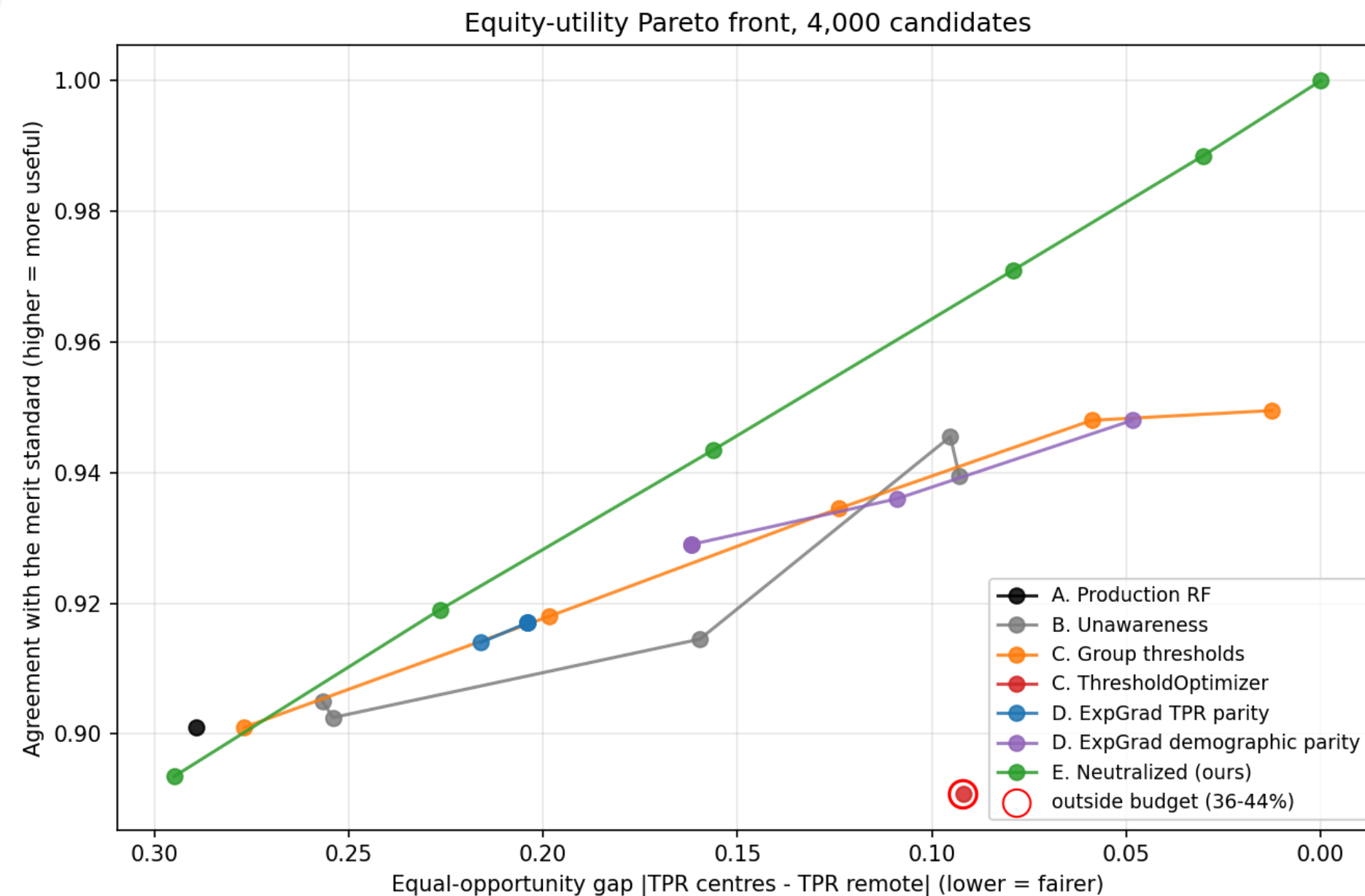
3

Rank

Score = $R + 0.145 \times \text{hours}$. Grant if ≥ 30.025 : 1,613 of 4,000 (40.3%), within the 36–44% budget.

One rule for everyone: no quota, no group thresholds, explainable to every applicant.

It beats every alternative we trained



fairlearn equal opportunity stalls at a 0.20 gap:
it equalizes against the biased labels.

Per-group thresholds close the gap only with
an explicit regional quota.

Dropping proxies also drops merit (hours).

Ours: gap 0.00, one rule for everyone.

Fairer and more accurate

On the 4,000 candidates	Production	Ours
Grant rate, centres	46.8%	40.4%
Grant rate, remote regions	27.8%	40.2%
Equal-opportunity gap (vs merit)	0.29	0.00
Accuracy vs hidden reference	89.33%	94.65%
Overall grant rate (budget 36–44%)	39.1%	40.3%

Fairness did not cost accuracy: removing the bias made the decisions **more** accurate.

Hidden reference = HxBuddy · ≈94.7% is the noise ceiling: no formula variant we tested scored higher

Keeping it fair in production

Monitor

Each round: parity gap above 5 pts triggers review. Blind human review of 200 files: EO gap above 0.05 suspends the model. Proxy drift (PSI above 0.2), per-region and intersectional gaps. Budget checked before publication.

Recourse

Every applicant sees their score and the cutoff. Human appeal within 30 days, with room for context. Quebec Law 25: notice of automated decision and the right to human review.

Guardrails

Never retrain on raw committee decisions. The list of merit vs context variables is frozen; changes need the fairness committee, including regional students. Annual external audit, model card, versioning.

One rule for everyone.

LIMITS WE OWN

The merit definition is a value judgement, made transparently and checked against the hidden reference. Hours worked partly reflect economic need. Two groups only: monitor each region and intersections.

[audit_rapport.ipynb](#) · [model_corrige.ipynb](#) · [predictions.csv](#)